Land-cover Change Detection

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Land-cover Change Detection

Objectives:

In this lab, we will explore a remote sensing application to create thematic land-cover maps, which will then be used for land-cover change detection. This process requires image classification, which involves converting multi-band raster imagery into a single-band raster with multiple categorical classes representing different land-cover types.

Background:

There are two primary methods for classifying a multi-band raster image: **Supervised** and **Unsupervised classification**. In supervised classification, the image is categorized based on spectral signatures (reflectance bands) obtained from training samples (points/polygons that represent distinct areas of the different land-cover types). You collect these samples, and the modeling algorithm uses them to classify the image.

In unsupervised classification, the algorithm identifies spectral classes (clusters) in the multi-band image without the analyst's intervention or providing any training samples. Once the clusters are created, you must determine what each cluster represents. For example, water, bare earth, or dry soil.

In summary:

Supervised: You collect training samples, provide them to the algorithm, and the algorithm learns from these samples to classify the image. Some example supervised classification algorithms are Maximum Likelihood, Random Trees (also called **Random Forests**), Support Vector Machine (**SVM**), and **K-Nearest Neighbor**.

Unsupervised: You instruct the algorithm to classify the image into a set number of classes without providing training samples. **ISO Cluster** is an example of an unsupervised classification algorithm.

In this lab, we will use supervised classification with the Random Forests algorithm. We have two aerial images of a part of Gainesville, one taken in 2008 and the other in 2023. These images are from **LABINS** **(http://www.labins.org/)** (Florida Department of Environmental Protection, FDEP). First, we will perform land-cover classification and then conduct land-cover change detection. These are the land-cover classes used in this lab: Water, Developed, Barren, Forest, and Herbaceous.

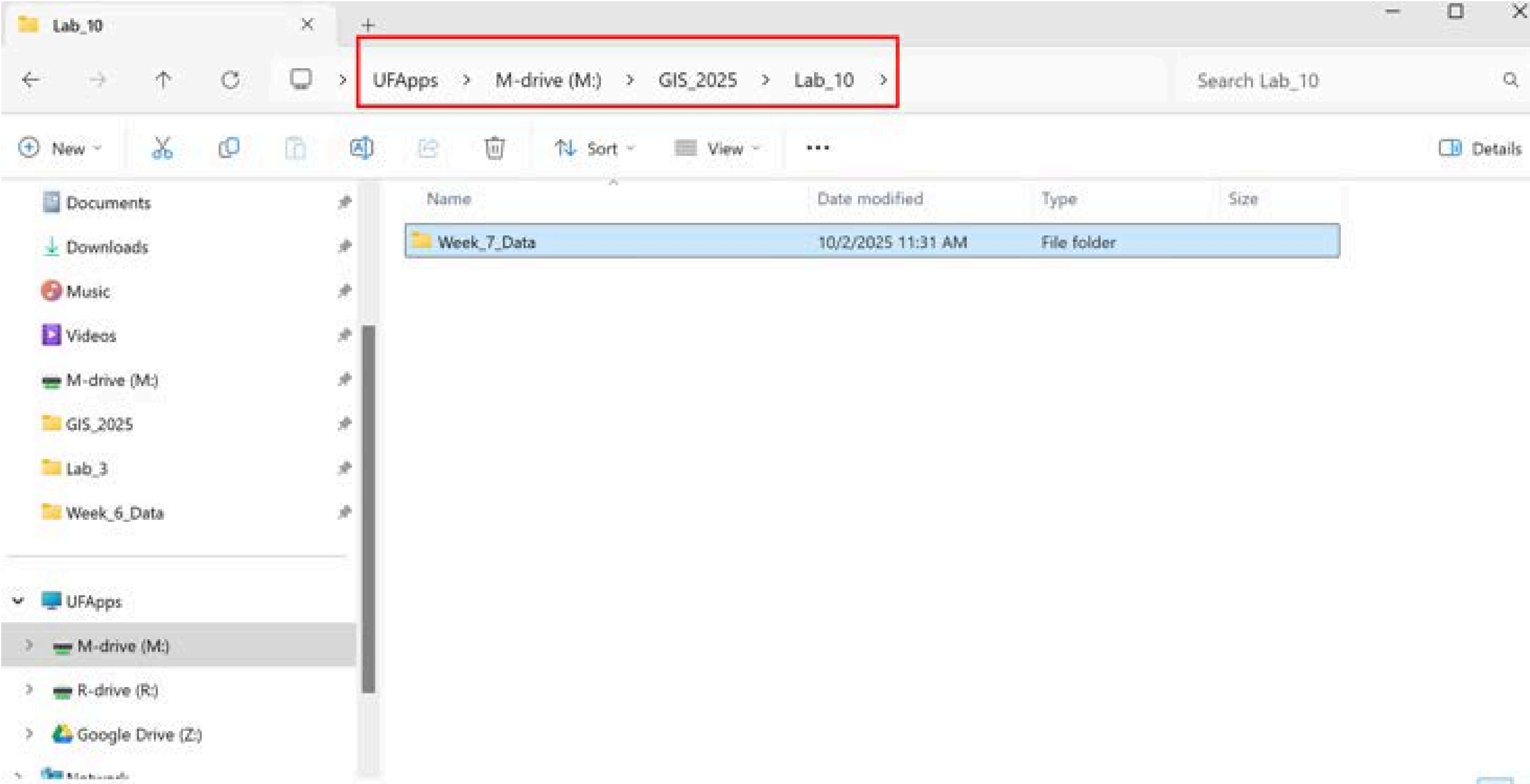
[Sometimes, this type of work is called land-use or land-use/land-cover (LULC) mapping. Land-use and land-cover are sometimes used interchangeably but they are different. Land-cover is what you can see on the ground, like water, trees, grass, or buildings. Land-use is about how people use the land, such as farming, living, or recreation. For example, a land-cover class like grass could have different land uses, such as being a ranch or a soccer field. In this lab, we will use the term land-cover]

(http://www.labins.org/)

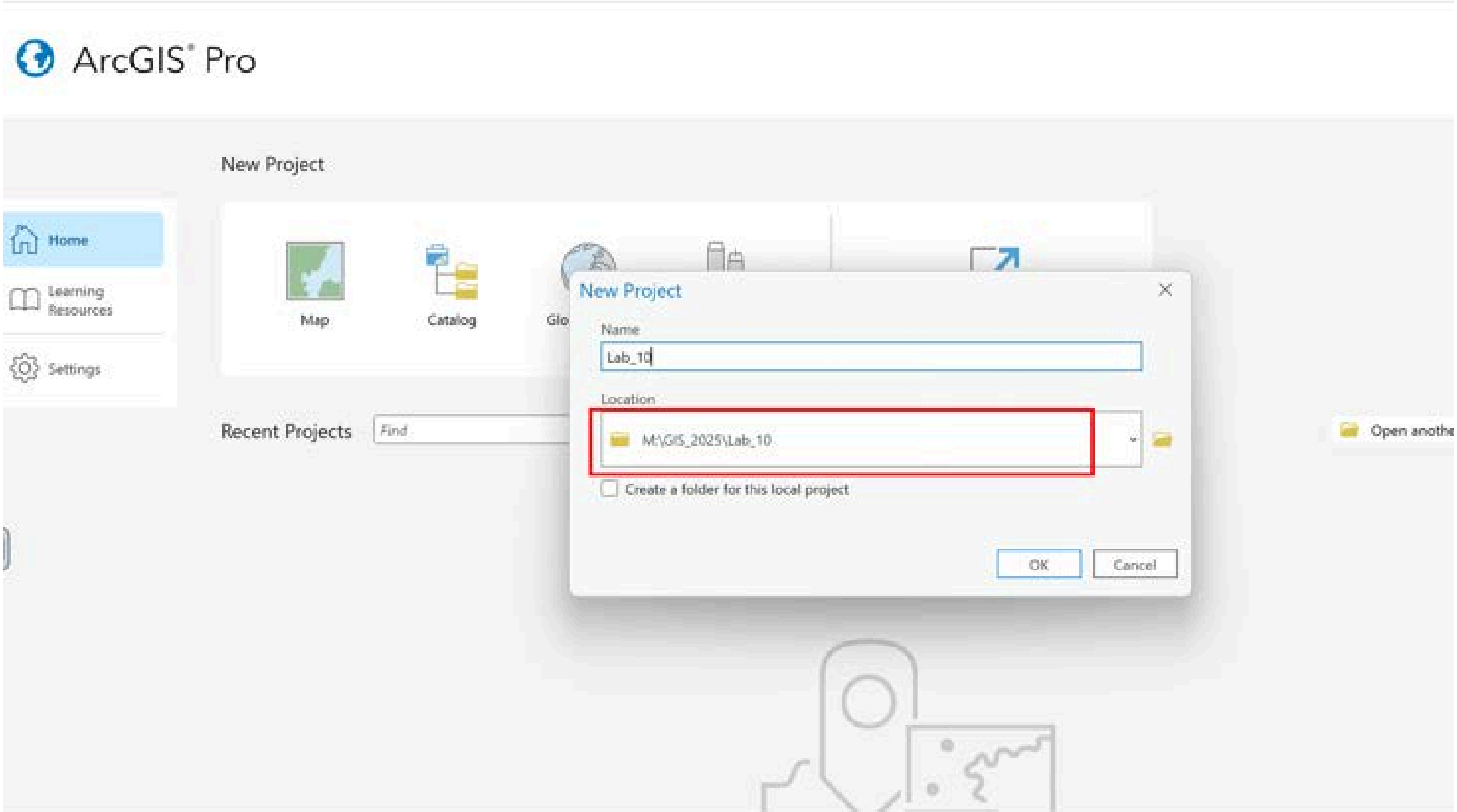
Steps:

1- Download the data, unzip the folder, and place it in your lab directory for use in ArcGIS Pro.

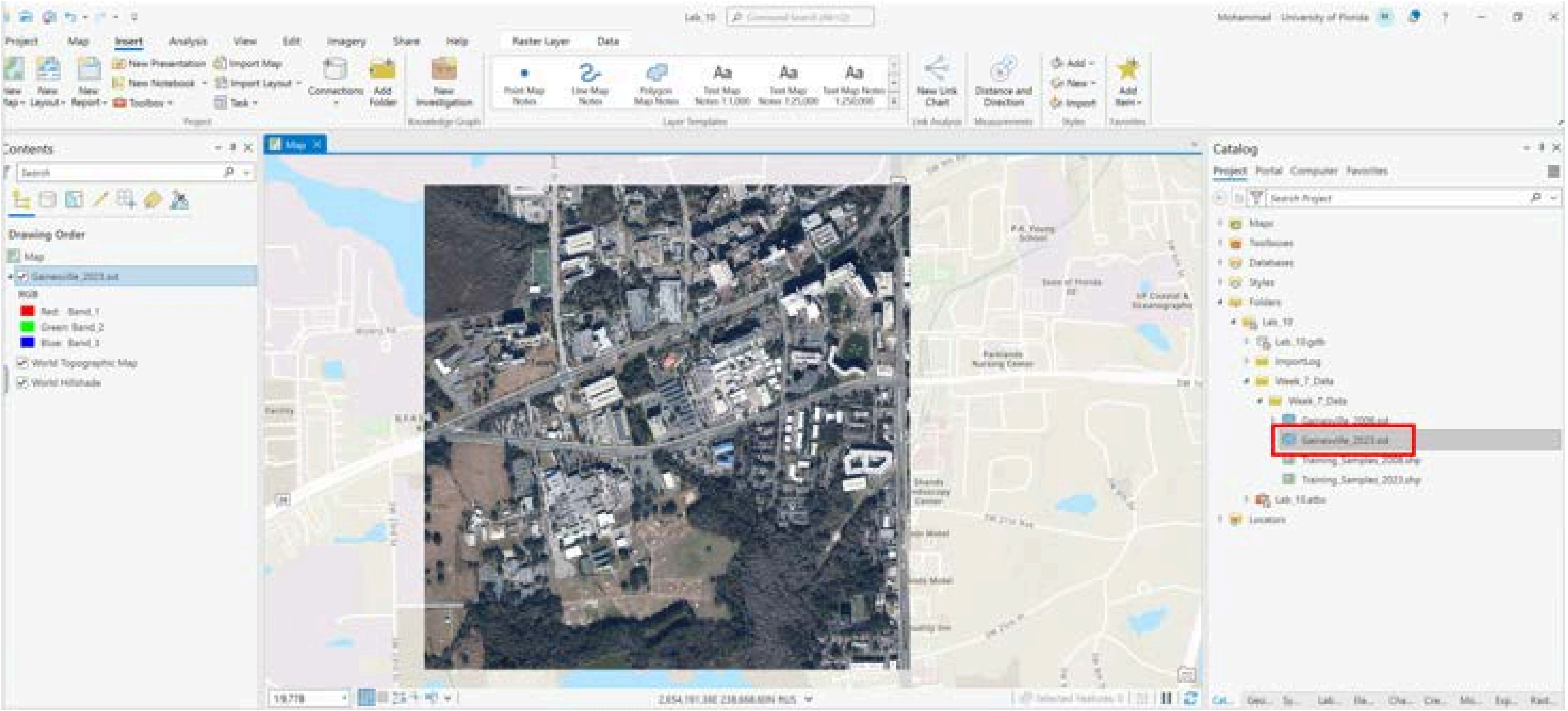
Week_7_Data.zip **(https://ufl.instructure.com/courses/541369/files/100665446?wrap=1)**
(https://ufl.instructure.com/courses/541369/files/100665446/download?download_frd=1)



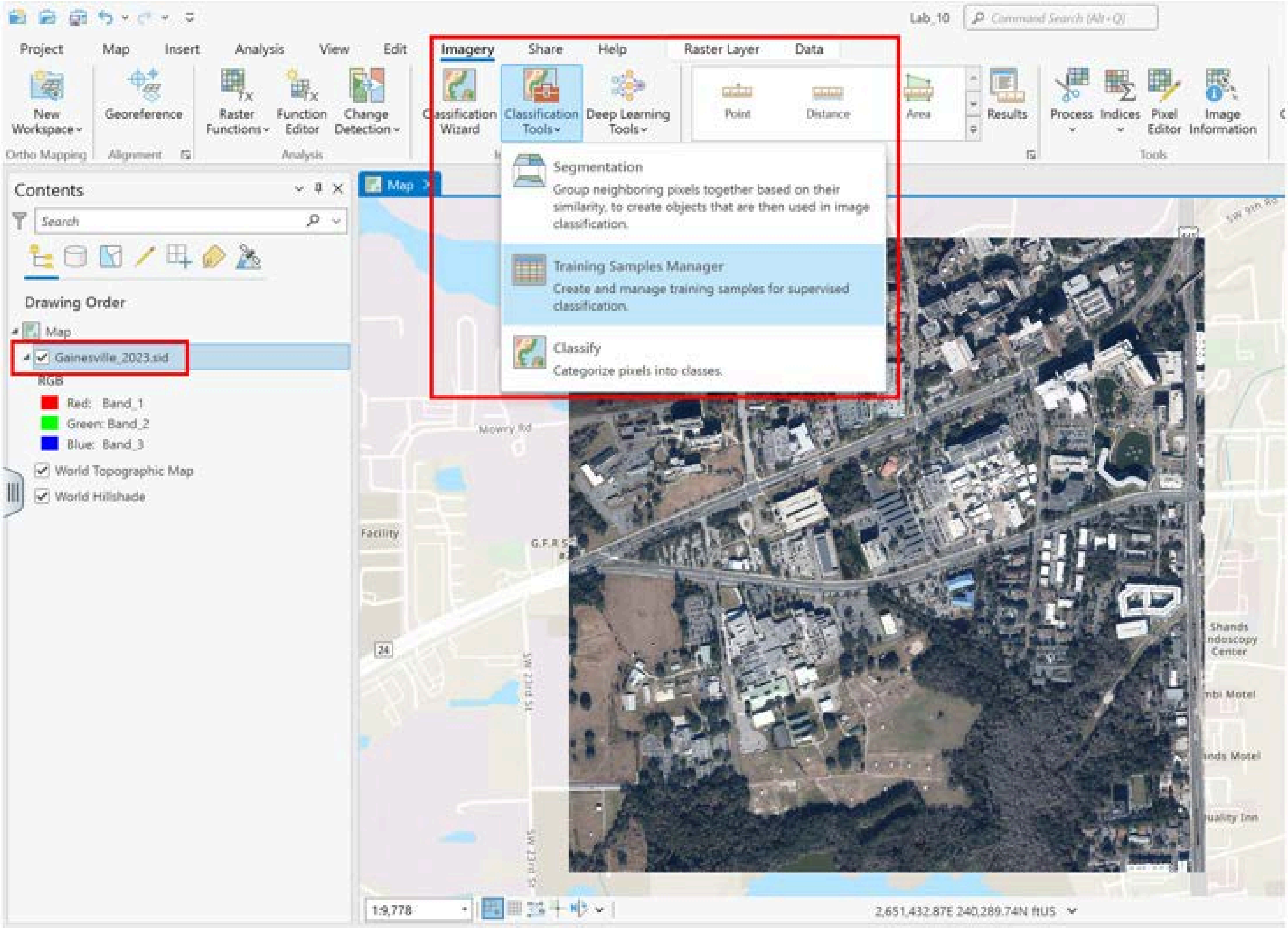
2- Open ArcGIS Pro, create a new map project, and save the project file in your lab data folder.



3- Add the Gainesville_2023.sid image to your map by dragging it into the Map view or using Map > Add Data > Browse.



4. Select Gainesville_2023.sid in the Contents pane, then go to Imagery > Classification Tools > Training Samples Manager.



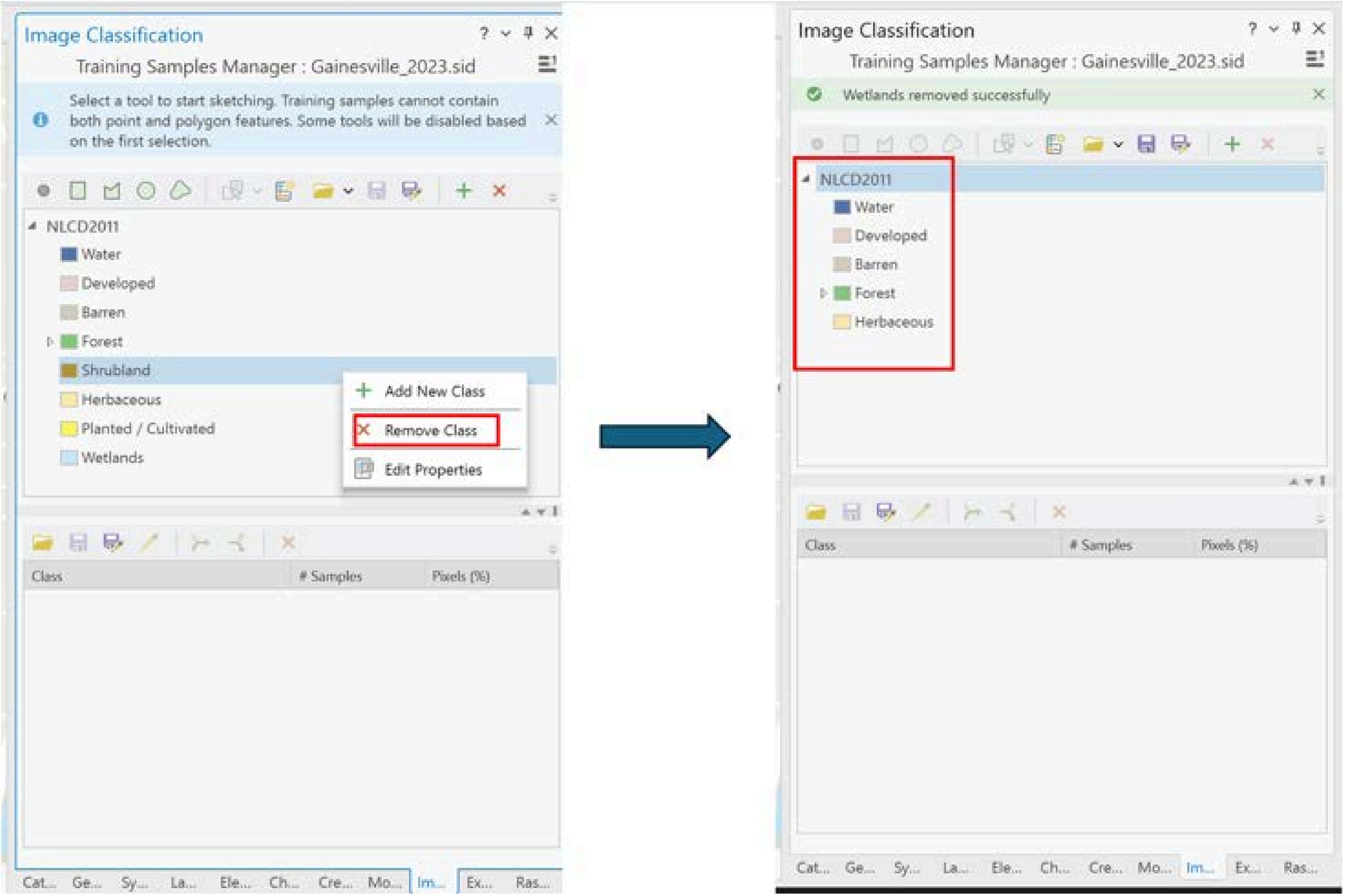
Lab_10
Command Search (Alt+Q)
Project
Map
Insert
Analysis
View
Edit
Imagery
Share
Help
Raster Layer
Data
New Workspace
Georeference
Raster Functions
Function Editor
Change Detection
Classification Wizard
Classification Tools
Deep Learning Tools
Point
Distance
Area
Results
Process
Indices
Pixel Editor
Image Information
Ortho Mapping
Alignment
Analysis
Tools
Segmentation
Group neighboring pixels together based on their similarity, to create objects that are then used in image classification.
Training Samples Manager
Create and manage training samples for supervised classification.
Classify
Categorize pixels into classes.
Contents
Search
Drawing Order
Map
Gainesville_2023.sid
RGB
Red: Band_1
Green: Band_2
Blue: Band_3
World Topographic Map
World Hillshade
Map
Mowry Rd
Facility
SW 23rd St
1:9,778
2,651,432.87E 240,289.74N ftUS

5. We want to classify the image into five land-cover classes:

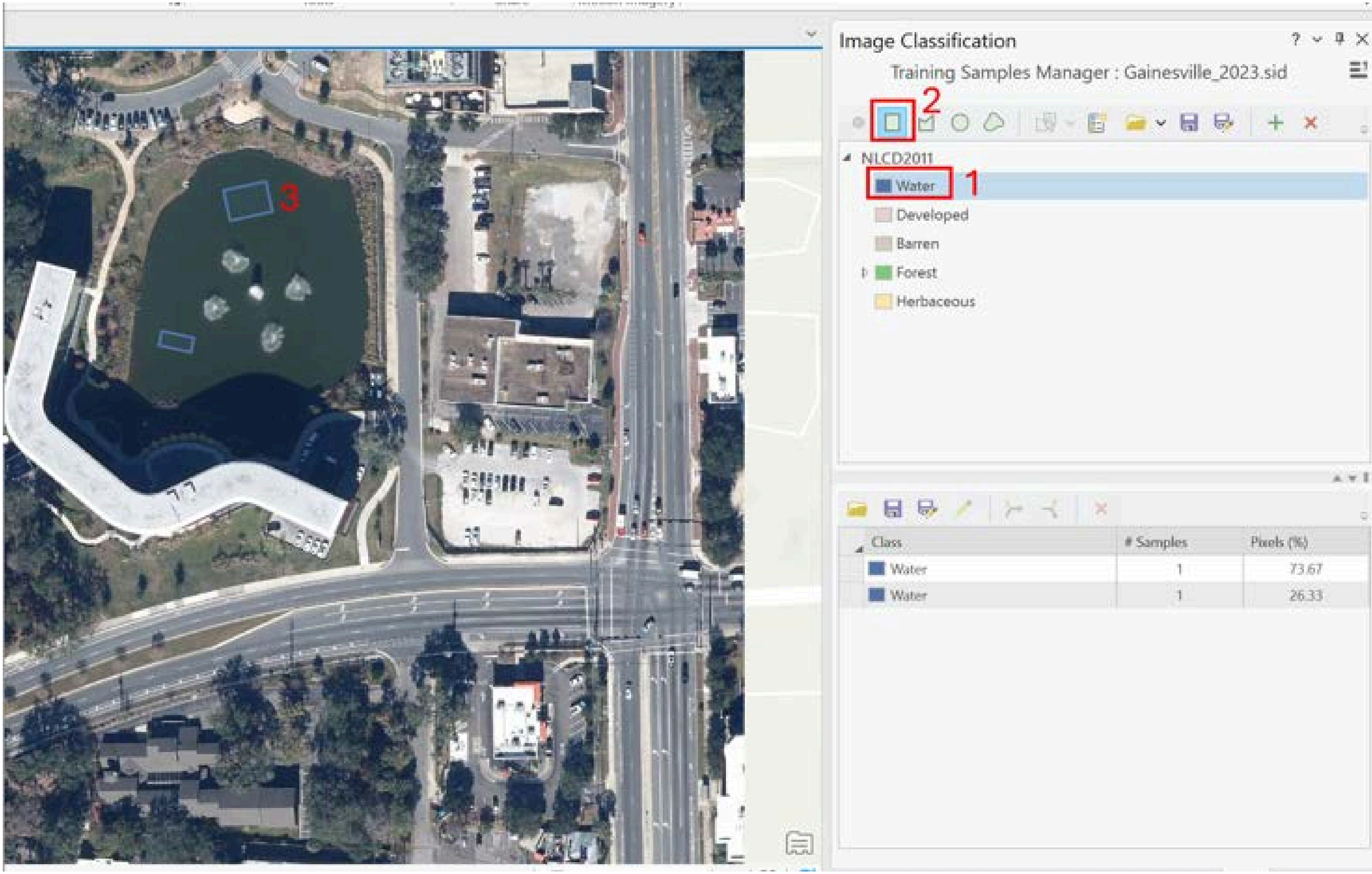
- Water
- Developed
- Barren
- Forest
- Herbaceous

You are welcome to collect your own training samples, but since this can take time, I have provided training samples in the Week_7_Data folder ("Training_Samples_2023.shp"), which you may use. If you want to collect your own samples, follow the steps below. Otherwise, skip to step 10.

6. Remove any classes other than these five by right-clicking on them and selecting Remove Class.



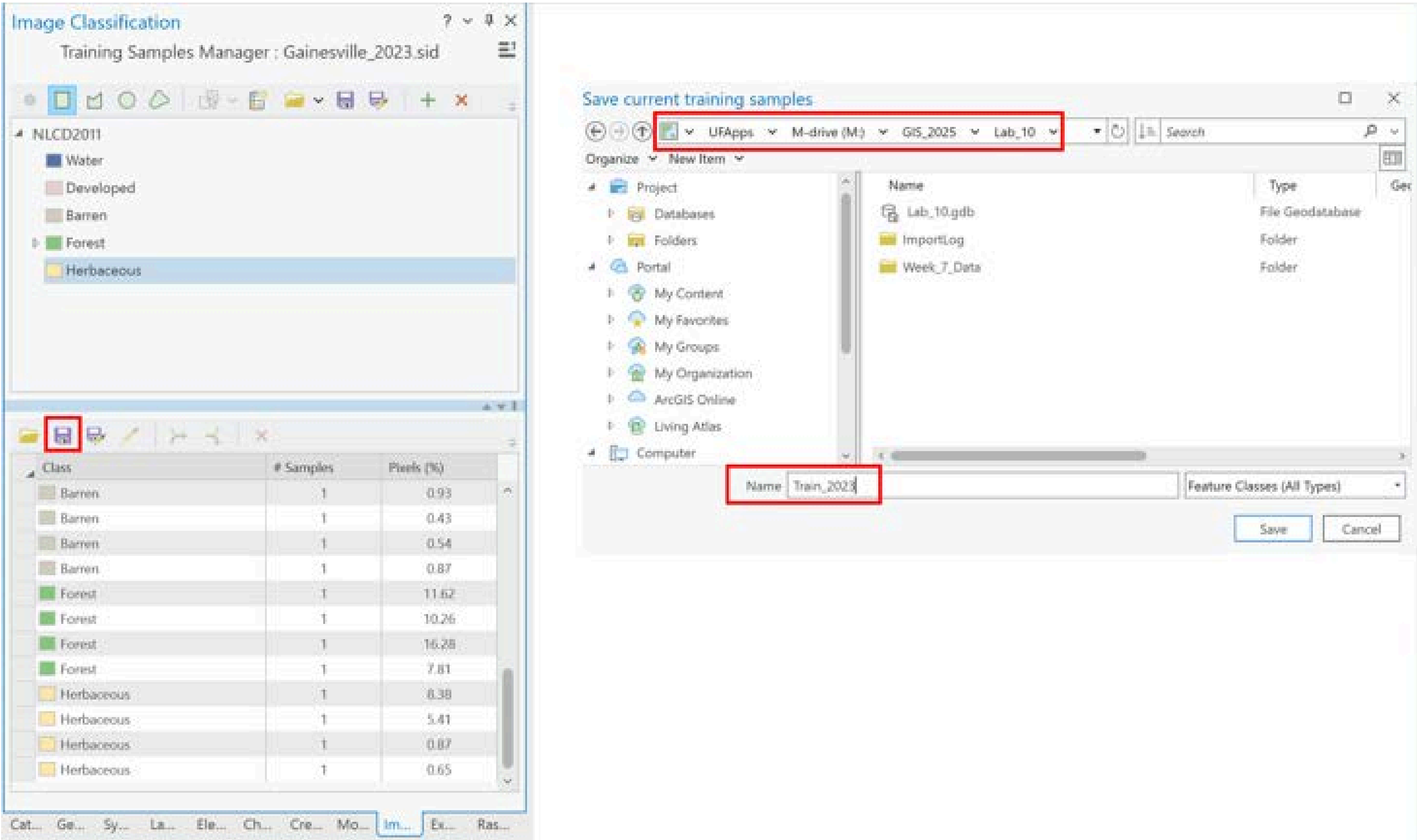
7. First, collect training samples for the Water class. Select Water, select polygon icon, then draw polygons on the water areas in the image.



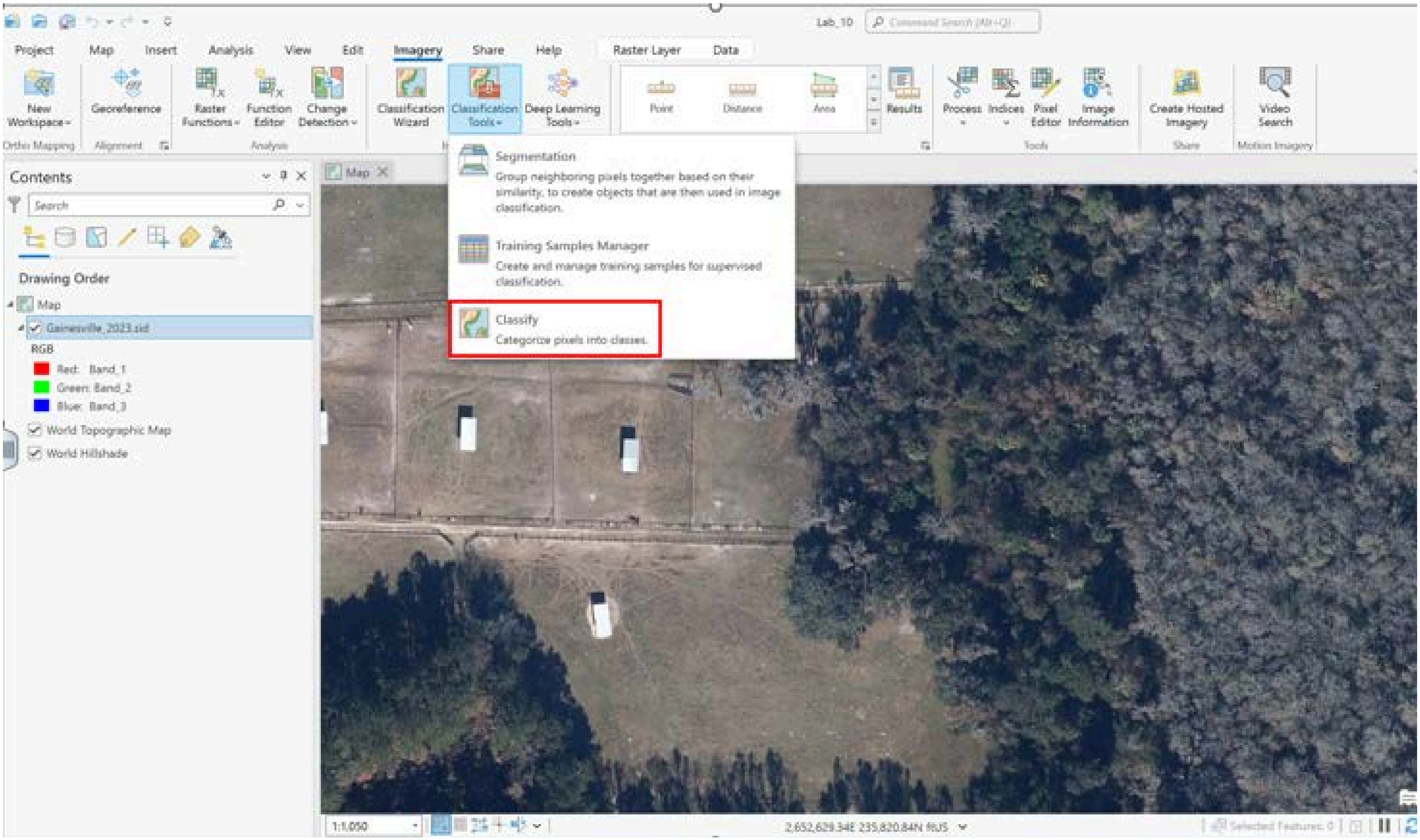
8. Repeat this step to collect polygon training samples for the other four land-cover classes.

While there are no exact guidelines on sample size, collecting more samples generally leads to greater classification accuracy.

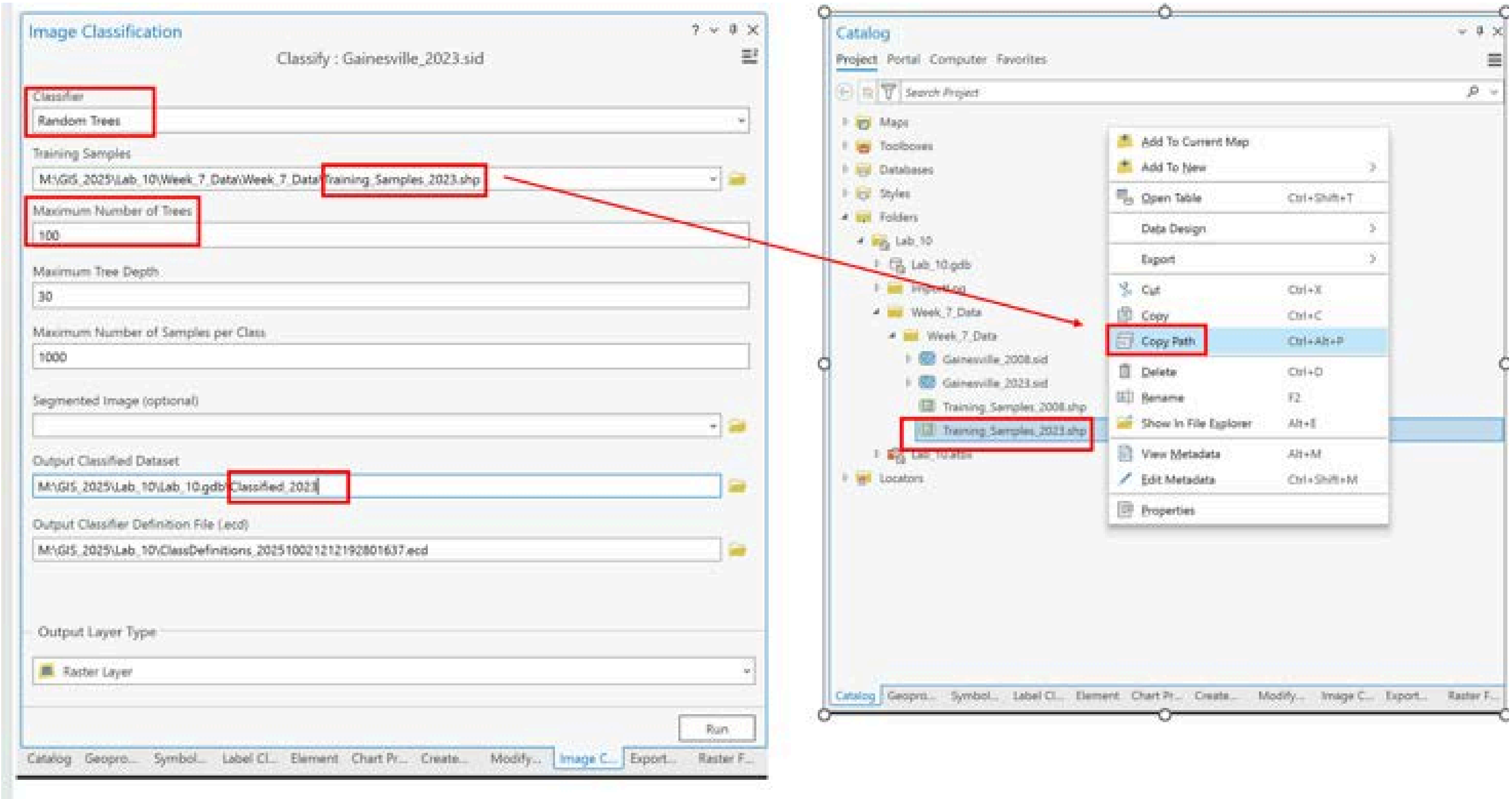
9. After collecting samples for all classes, select the Save icon and save the file in your lab folder with the name Train_2023.



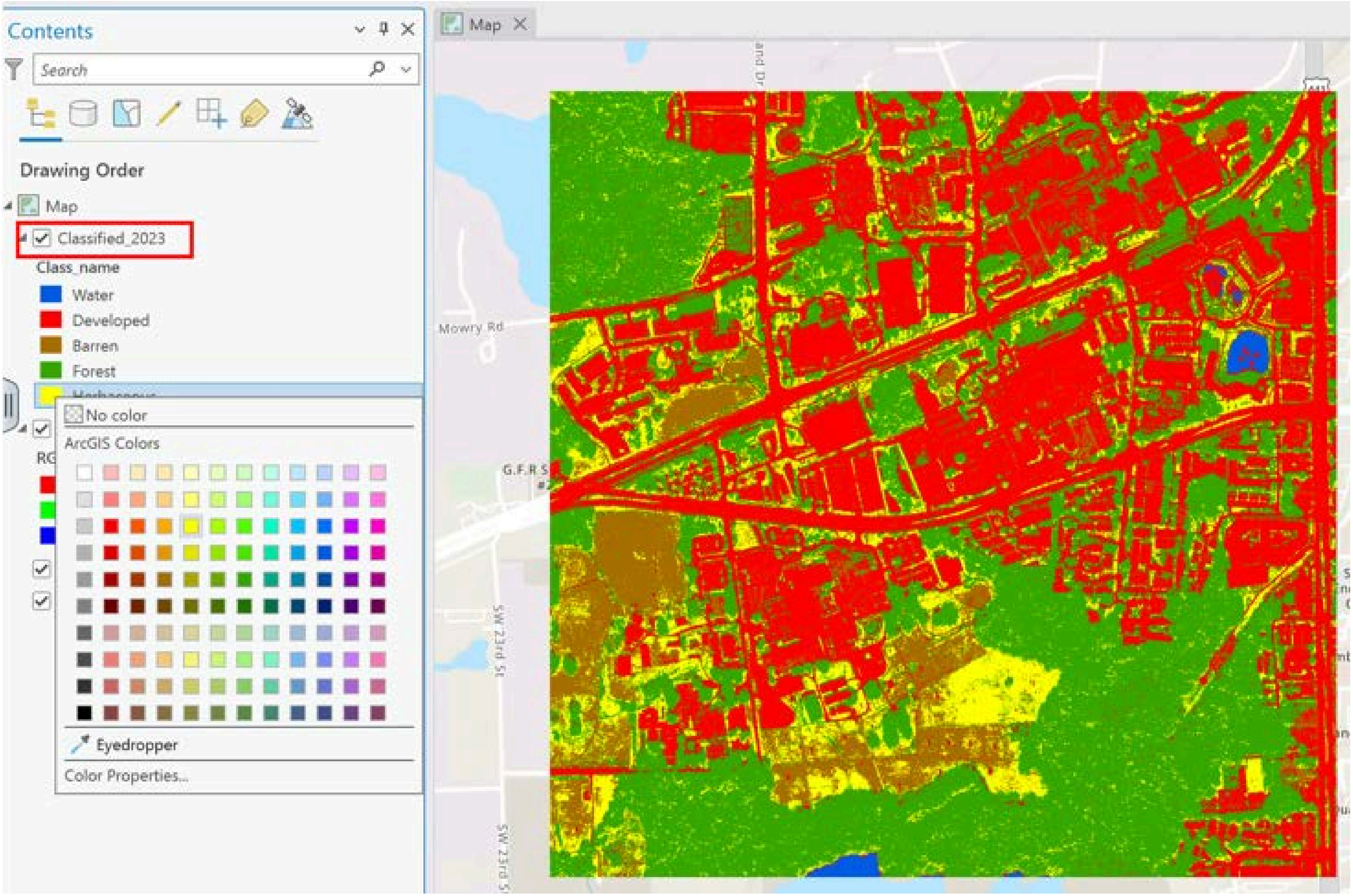
10. Now that you have collected your training samples, we want to train a model to learn from these samples and classify the entire image into the five land-cover classes. There are several models available; in this lab, we will use the Random Forests algorithm. Go to Classify in the Imagery tab.



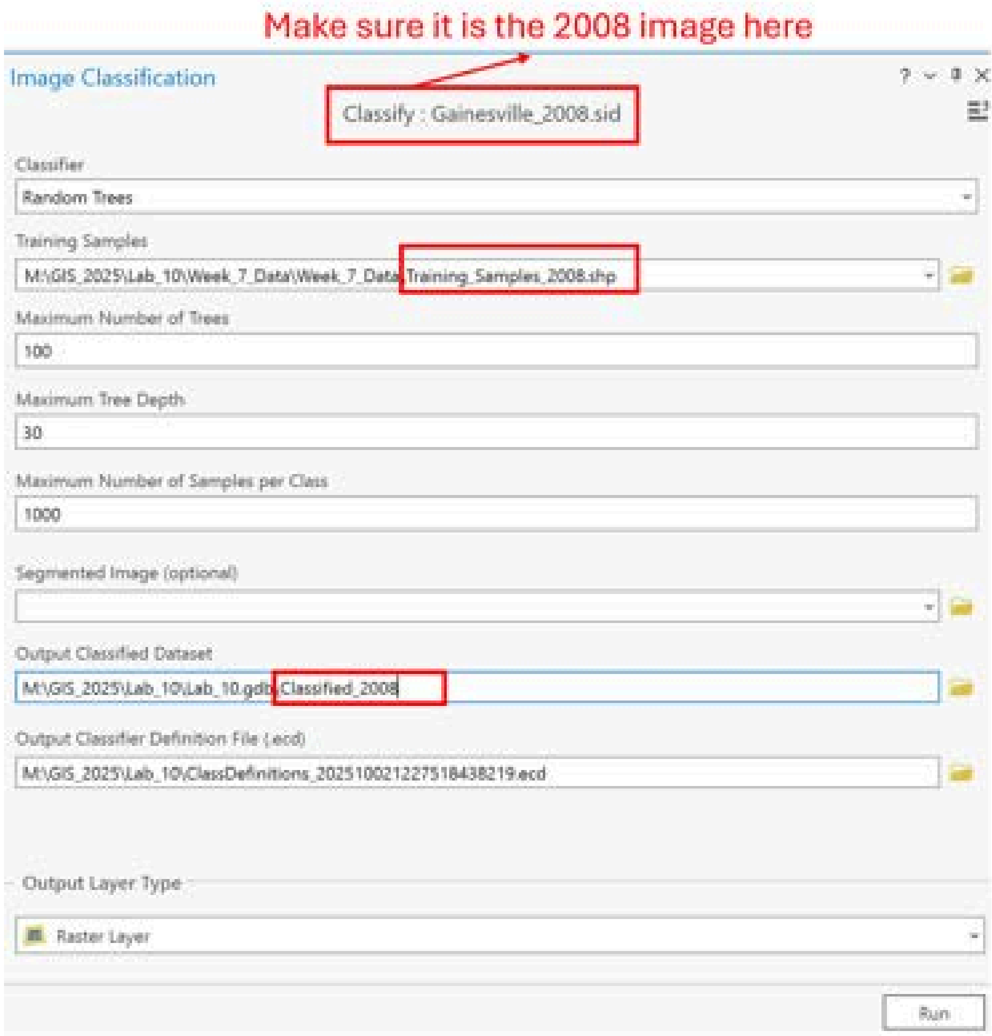
11. Set the following options:
    Classifier: Random Trees
    Training Samples: Copy and paste the path of the training samples you created or the one I provided
    Maximum Number of Trees: 100
    Output Classified Dataset: Classified_2023
    Leave the other options as default and click Run.

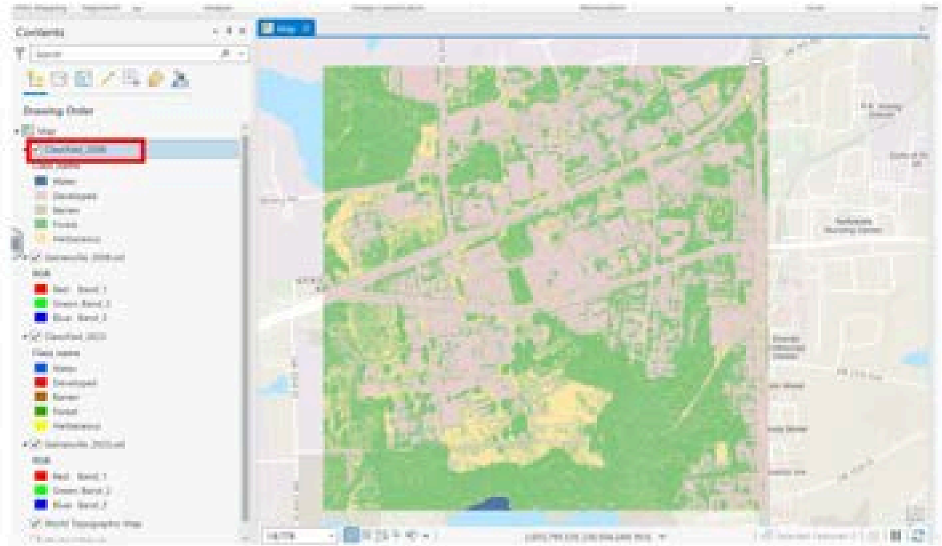


12. This will classify all the pixels in the image and assign each pixel to one of the five land-cover classes. You can change the color of each class by right-clicking on its color in the Contents pane.

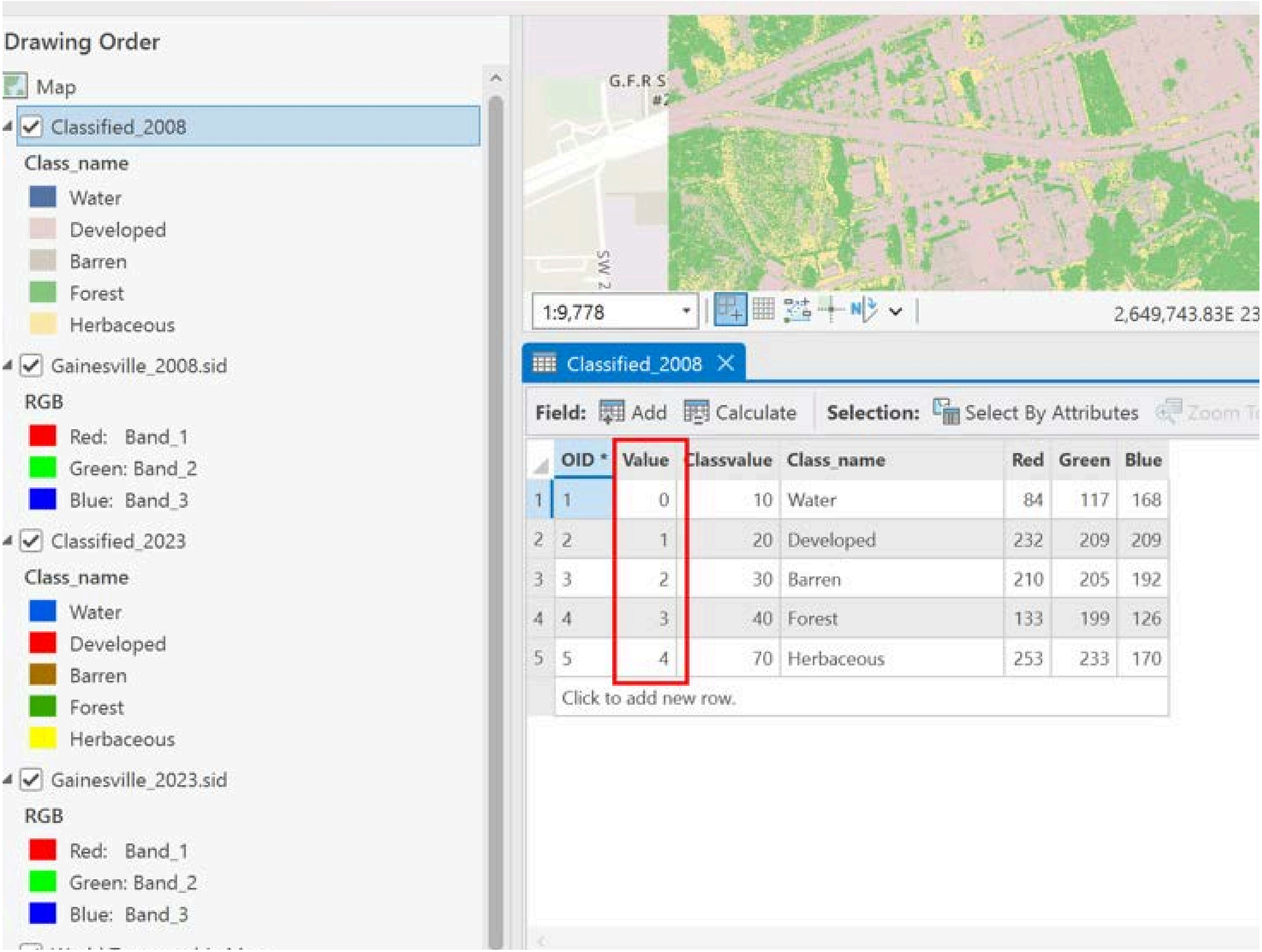


13. Repeat steps 3–12 using Gainesville_2008.sid. You can collect your own training samples or use the ones I provided (Training_Samples_2008.shp).



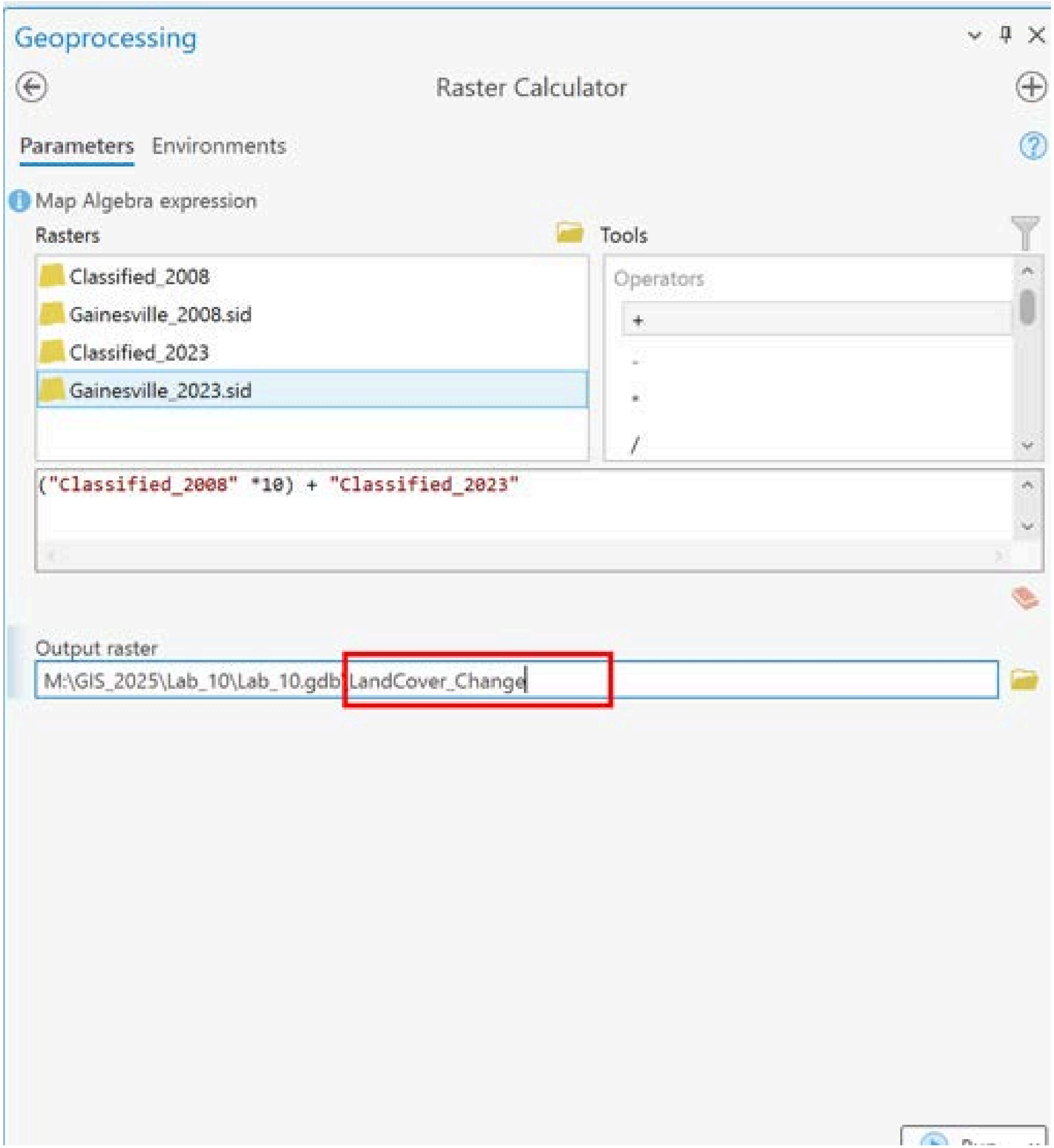


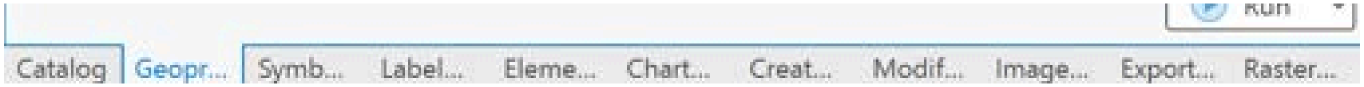
14. If you look at the attribute tables of the classified images, you will see that the "Value" field has a unique number (0 to 4) for each land-cover class.


Drawing Order
Map
Classified_2008
Class_name
Water
Developed
Barren
Forest
Herbaceous
Gainesville_2008.sid
RGB
Red: Band_1
Green: Band_2
Blue: Band_3
Classified_2023
Class_name
Water
Developed
Barren
Forest
Herbaceous
Gainesville_2023.sid
RGB
Red: Band_1
Green: Band_2
Blue: Band_3
1:9,778
2,649,743.83E 23
Classified_2008
Field: Add Calculate
Selection: Select By Attributes
OID * | Value | Classvalue | Class_name | Red | Green | Blue
1 | 1 | 0 | 10 | Water | 84 | 117 | 168
2 | 2 | 1 | 20 | Developed | 232 | 209 | 209
3 | 3 | 2 | 30 | Barren | 210 | 205 | 192
4 | 4 | 3 | 40 | Forest | 133 | 199 | 126
5 | 5 | 4 | 70 | Herbaceous | 253 | 233 | 170
Click to add new row.

15. Now we want to detect land-cover changes from 2008 to 2023 using "from-to" encoding, which captures all possible types of change between the classes. From Analysis > Tools, search for Raster Calculator (Image Analyst Tools). Type the following expression and save the output raster as LandCover_Change.

("Classified_2008" *10) + "Classified_2023"


Geoprocessing
Raster Calculator
Parameters
Environments
Map Algebra expression
Rasters
Tools
Classified_2008
Gainesville_2008.sid
Classified_2023
Gainesville_2023.sid
Operators
+
-
*
/
("Classified_2008" *10) + "Classified_2023"
Output raster
M:\GIS_2025\Lab_10\Lab_10.gdb\LandCover_Change



When we combine the 2008 and 2023 classified rasters using this operation, a new raster image will be created. Each pixel in this image will be assigned a two-digit code.

- **The first digit** = land cover class of that pixel in **2008**.
- **The second digit** = land cover class of that pixel in **2023**.

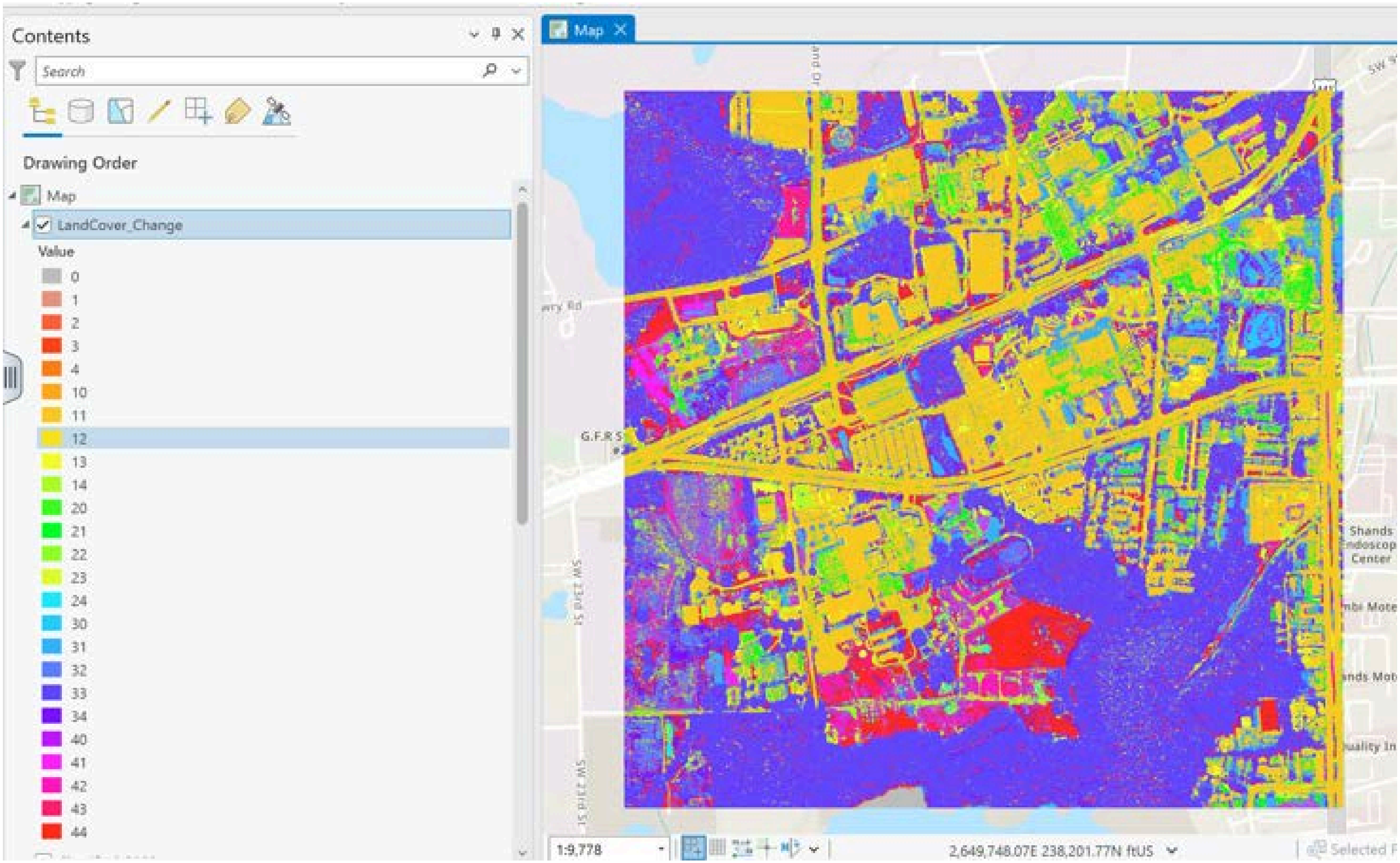
This lets us see **from–to changes** in a compact way.

Our class codes were as follow:

- 0 = Water
- 1 = Developed
- 2 = Barren
- 3 = Forest
- 4 = Herbaceous

In the LandCover_Change output image, each pixel gets a two-digit code that shows the change from 2008 to 2023. For example:

- **0 (i.e., 00)** → Pixel stayed **Water (0)** in both 2008 and 2023 (**no change**).
- **1 (i.e., 01)** → Pixel was **Water (0)** in 2008 → became **Developed (1)** in 2023.
- **10** → Pixel was **Developed (1)** in 2008 → became **Water (0)** in 2023.
- **11** → Pixel stayed **Developed (1)** in both 2008 and 2023 (**no change**).



Since we had five unique classes in each image, there are 25 possible "from-to" class changes between the two years.

16. Now we want to show only the top five main land-cover changes by their frequency in the study area. Open the attribute table of the LandCover_Change image and double-click the Count field twice to sort it in descending order. Identify the top five changes, excluding the no-change pixels (0, 11, 22, 33, 44).

Here, the following values were identified as the top five main land-cover changes:

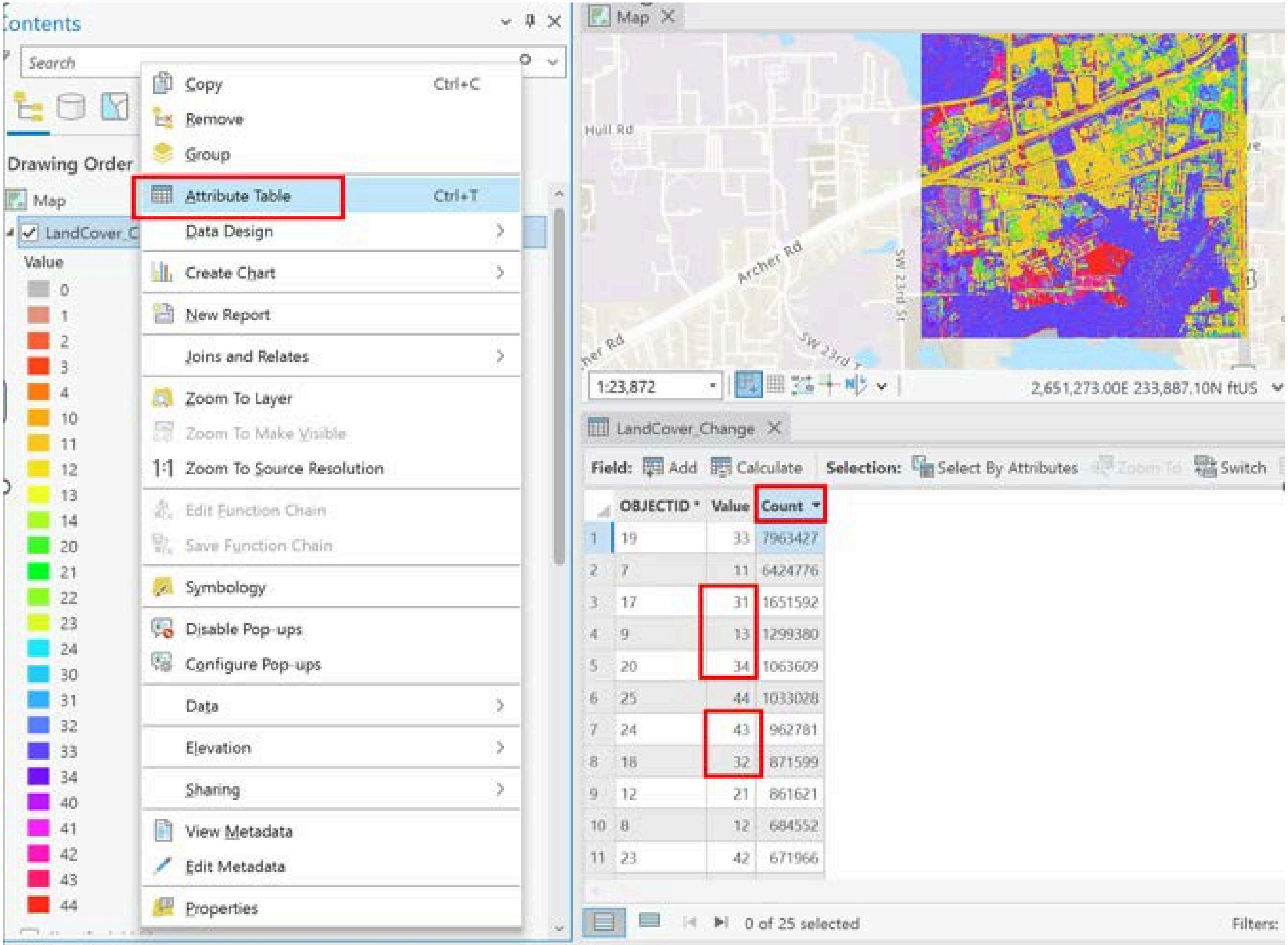
31: Forest to Developed

13: Developed to Forest

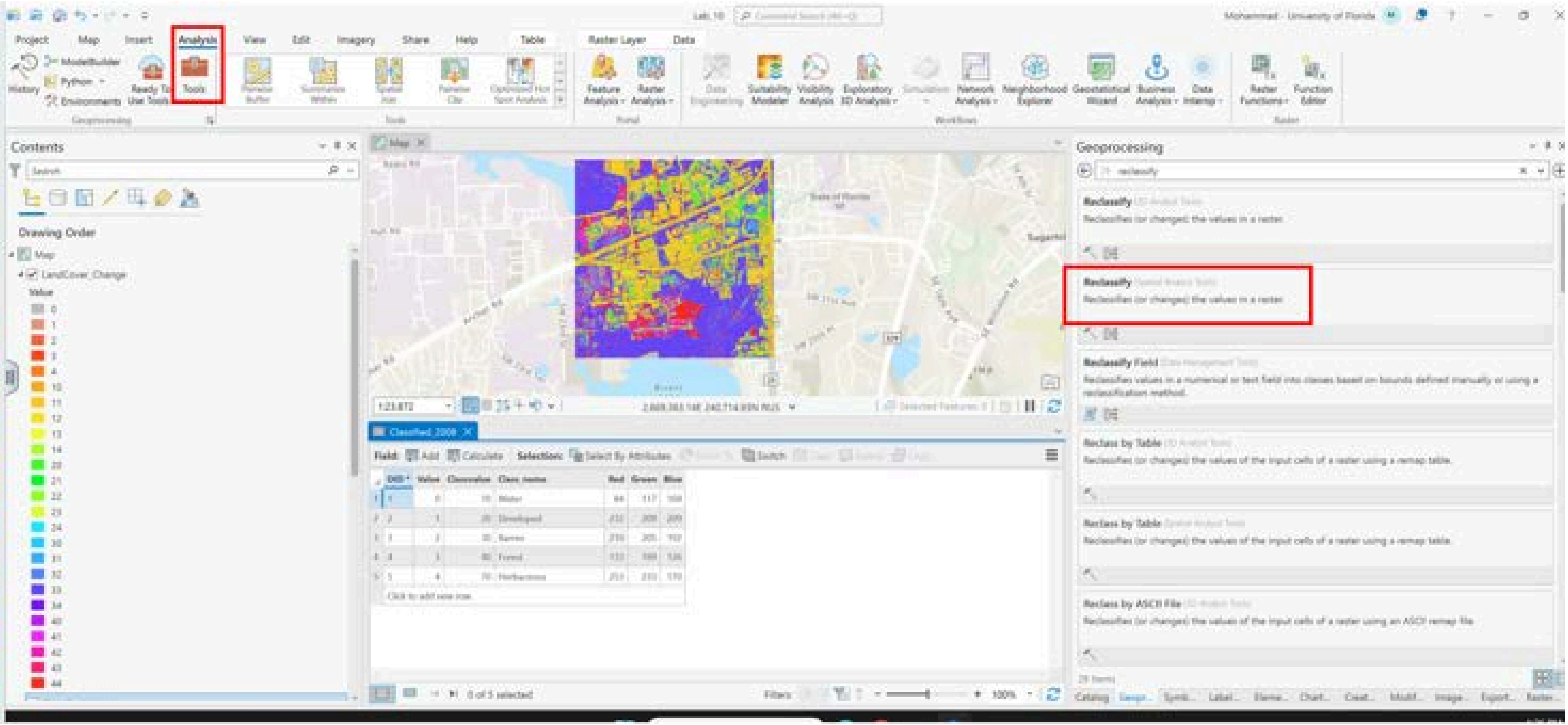
34: Forest to Herbaceous

43: Herbaceous to Forest

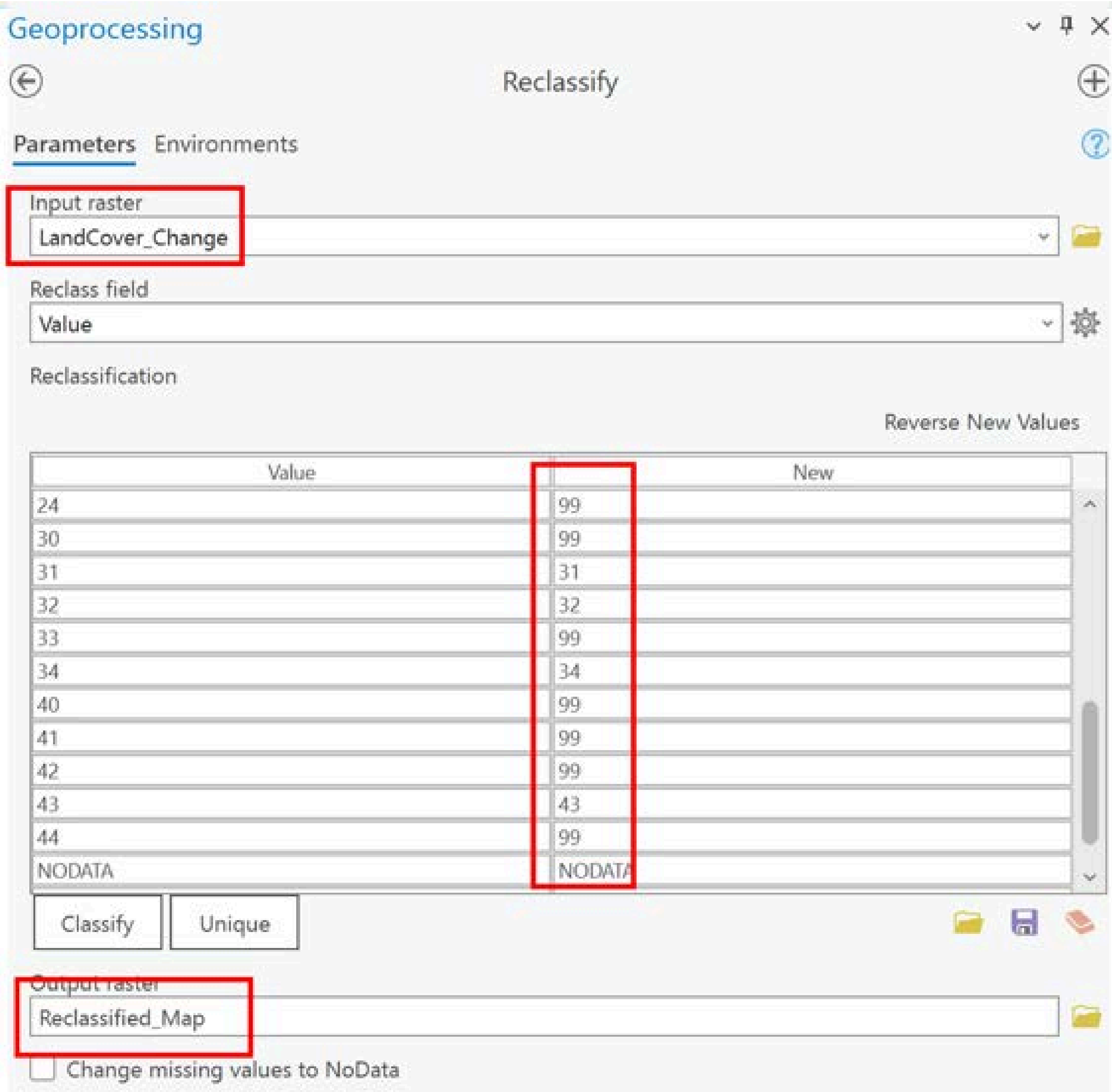
32: Forest to Barren

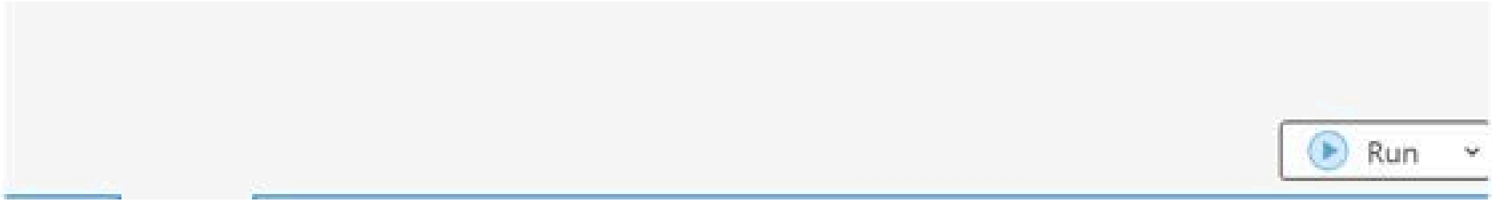


17. For all other values and for no-change pixels (00, 11, 22, 33, 44), we want to merge them into a single class called "no-change/not significant change." To do this, go to Analysis > Tools > Reclassify (Spatial Analyst Tools).

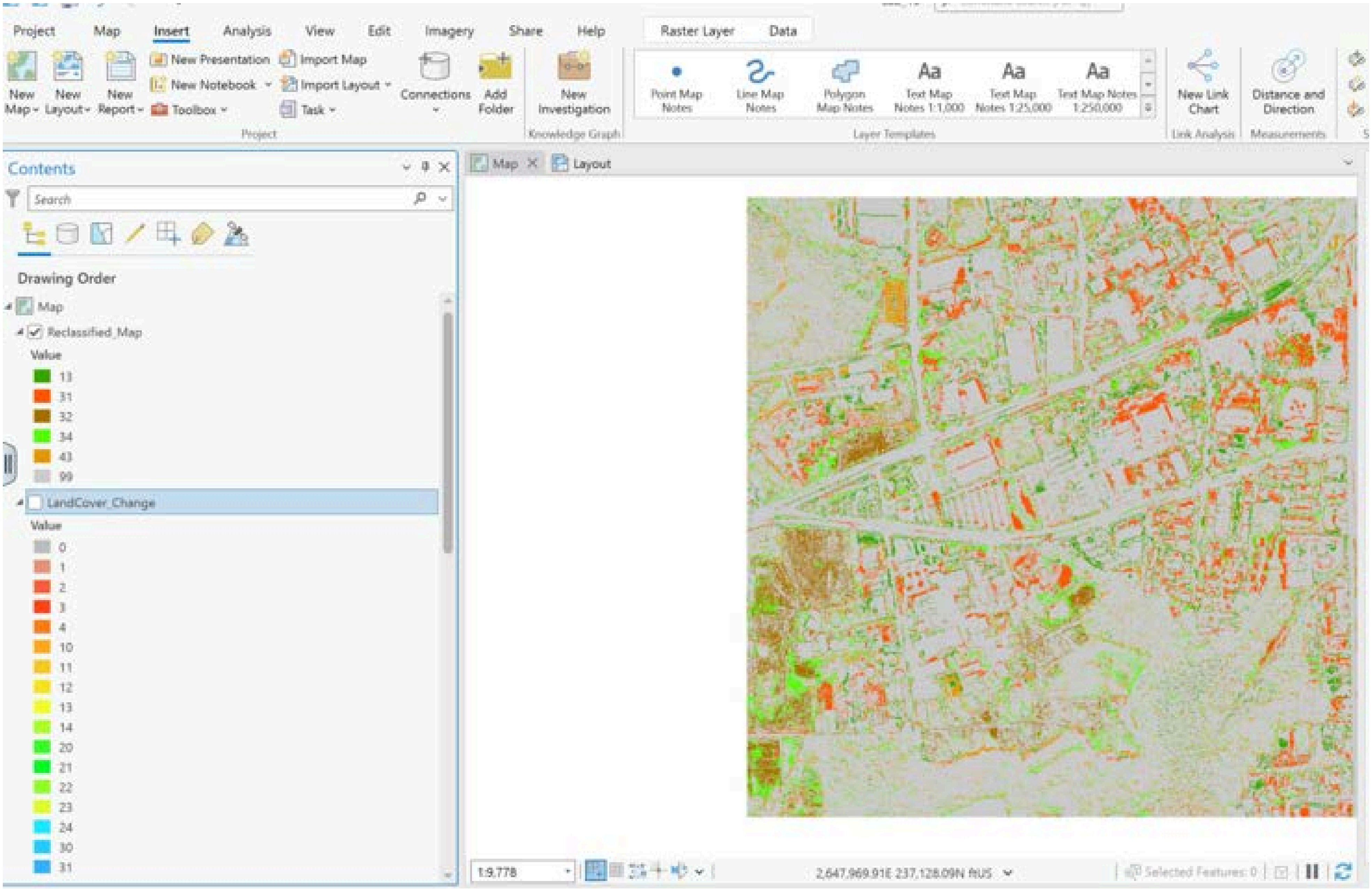


18- Give a unique New value (e.g. 99) to all changes other than the top 5 selected (31, 13, 34, 43, and 32). Name the output file Reclassified_Map and click Run.


Geoprocessing
Reclassify
Parameters
Environments
Input raster
LandCover_Change
Reclass field
Value
Reclassification
Reverse New Values
Value
New
24
99
30
99
31
31
32
32
33
99
34
34
40
99
41
99
42
99
43
43
44
99
NODATA
NODATA
Classify
Unique
Output raster
Reclassified_Map
Change missing values to NoData



19. Unselect all other layers in the Contents pane, and pick appropriate, contrasting colors for each change. I used grey for the no-change or not significant change class.



20- Make a Map Layout

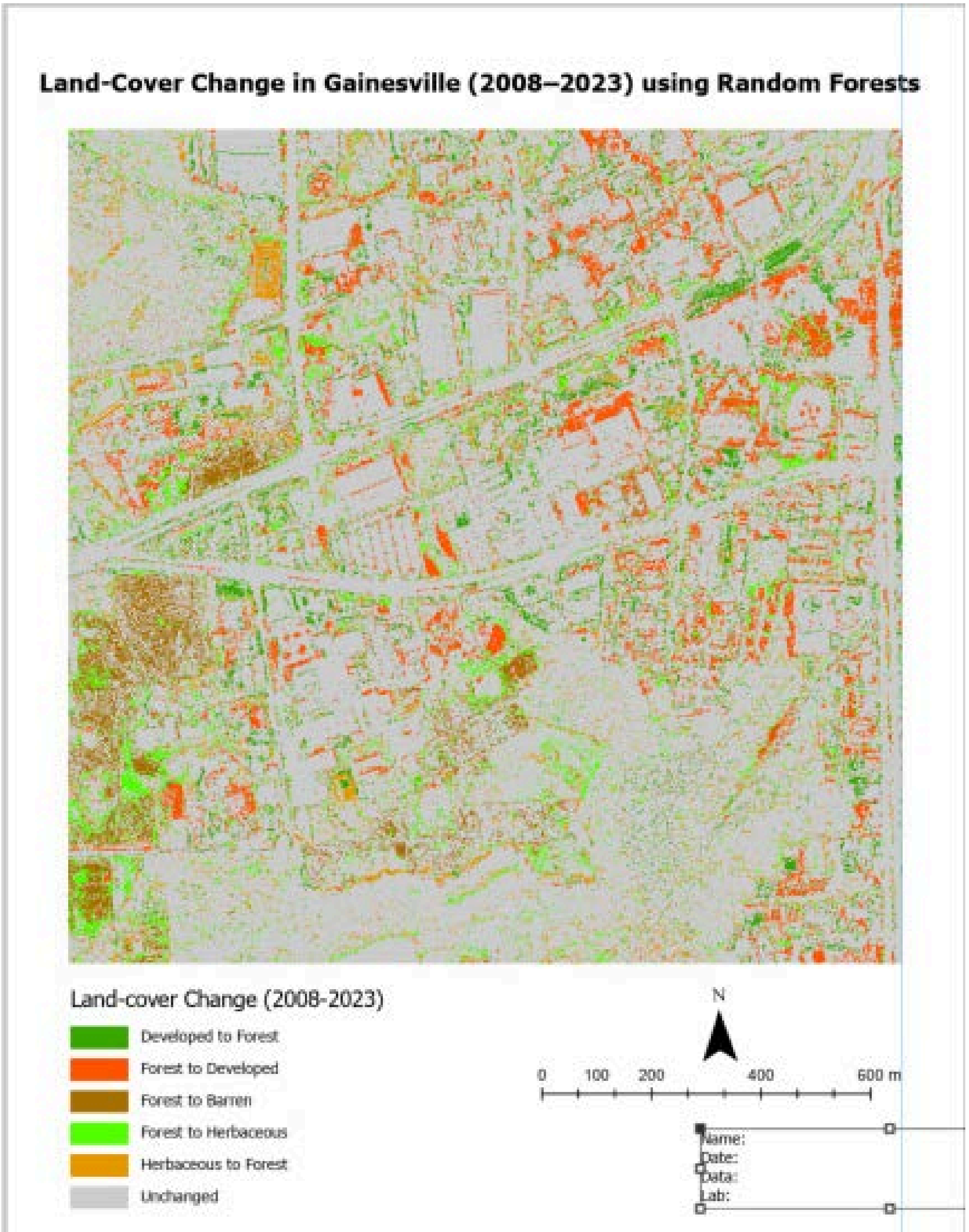
- From the Insert tab, create a new layout and add your map frame.
- Include a title, legend, non-invasive scale bar and north arrow, and your identification (name, date, data etc.) on the map.

- Export and submit your map as as a PDF (please do not submit a screenshot).

Checklist before submission:

Aesthetics and Map Elements

- The map frame clearly highlights the top five main land-cover changes and unchanged areas **(7 pts)**.
- Colors are high-contrast and visually appropriate **(1 pt).**
- Layout balance: map elements are well-sized and avoid excessive empty space **(1 pt)**.
- Required elements included: title, legend, scale bar, north arrow, identification (name, date, data, etc.) **(2 pts).**
- Legend quality: easily readable, well-sized, has a descriptive title, and free of unnecessary labels or the GIS layer names **(1 pt).**


Land-Cover Change in Gainesville (2008–2023) using Random Forests
Land-cover Change (2008-2023)
Developed to Forest
Forest to Developed
Forest to Barren
Forest to Herbaceous
Herbaceous to Forest
Unchanged
N
0
100
200
400
600 m
Name:
Date:
Data:
Lab:

Further tips to consider for future research:

In this lab, training samples were collected based on visual interpretation of the image, and we did not use ground truth data to verify the results. The goal was simply to demonstrate how classification works in ArcGIS Pro. For your own projects, you should always check your results by comparing your classified data with ground truth or other reliable sources.

Also, in this lab we used only a few training samples, just three RGB bands as variables, and ran the classification once with an arbitrary parameter value. For higher accuracy in practice, you should:

- Evaluate your classification both visually and quantitatively, using tools such as a confusion matrix, overall accuracy, or F1-score.
- Use a large number of training samples for better accuracy.
- Include more independent variables than just the RGB bands. Other useful variables include spectral indices (like NDVI or NDWI), terrain data, and textural features.
- Test different models by running your classification multiple times with various algorithms (such as SVM, K-Nearest Neighbor, or deep learning algorithms) and by tuning different parameters (for example, the number of trees in Random Forests) to find what gives you the best results.

Points	12
Submitting	a file upload

Due	For	Available from	Until
Oct 12	Everyone	Oct 6 at 12am	Oct 17 at 11:59pm

+ **Rubric**